

accessible to external entities.

Algorithms such as AES and DES are commonly used so the encryption benchmarks tell you whether you'll have a smooth workflow in such scenarios.

5. Single/Multi-threaded - Most of the apps running on your PC are single threaded and a high score in such benchmarks tell you how your day to day experience with your PC will be. A growing number of applications are switching to optimising their code so that it can run better on multi-core processors but the priority is still on single threaded performance.

HOW DO YOU COMPARE?

If you're looking at just the clock speed in GHz or the number of cores and threads then you'll only have a partial idea as to how the CPU is going to perform. Modern CPUs have many different components at the architectural level that are focused on improving

performance in certain workloads. Take for example the IGP (Integrated Graphics Processor) in Intel processors, it's good enough to play games with rudimentary graphics but they're also tied into certain features. HDR on Netflix will only be available if you have an Intel 7th Gen processor or an NVIDIA 1050 or higher. This is because the IGP has incorporated the Widevine DRM requirements.

So how can you quickly compare? To do that, you need to prioritise which tasks are going to form the bulk of your workload. If you perform video editing, then you need more cores. If you want to game, then you need fewer cores running at higher speeds. If you're going to multitask, then the software you'll be using should be optimised for more cores otherwise getting a HEDT processor won't offer the best value for money. This is why benchmarks exist and are broken down into so many different categories. If you want to run programming

software then opt for a processor that has a lot of cache. Massive programs can take forever to compile, in such cases it's beneficial to have higher clock speeds as well as more cores.

Simply put, there is no easy way to compare two CPUs because the "performance" is the culmination of multiple factors put together. Which is why we always ask folks to list the usage scenario so that a contextual performance comparison can be done.

A CLIFFHANGER!

CPU microarchitectures are extremely complex. We've tried our best to give you a broad overview and put down a few concise methods to determine how a CPU fares in a certain workload. However, if you're really interested in knowing how each processor functions right down to the bit level, then both Intel and AMD have white papers that explain quite a lot without going into the proprietary features. You might want to read those. 

GPU ARCHITECTURE EXPLORED

RELATIVELY SIMPLER
THAN CPUS BUT A
LOT MORE POWERFUL

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You've no doubt figured out how CPUs work to help crunch out all the many types of operations that come together to form a CPU. The GPU or the Graphics Process-

ing Unit works in somewhat a similar manner. However, the type of calculations performed by the GPU form a fraction of what the CPU is capable of doing. Over the years, the graphics card has evolved to be more specialised into not only a graphics accelerator but also for new technology domains such as AI. Most of which was made possible thanks to the concept of the unified

shader architecture that was accepted by both, AMD and NVIDIA. While the microarchitectures do still differ from each other, graphics cards have ended being being capable of doing much more. Before we get into how AMD and NVIDIA microarchitectures differ from each other, let's try to understand how the graphics pipeline functions. Soon enough we'll be looking at another major com-